

ISLAND LAKE, MB, Canada

WMO: 711450

Lat: 53.854N

Lon: 94.653W

Elev: 773

StdP: 14.29

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.1	-28.9	-41.1	0.5	-33.0	-37.0	0.7	-28.7	19.0	-0.9	17.1	-0.7	3.8	300	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	82.2	66.5	79.1	64.3	76.1	62.9	69.0	78.1	67.0	75.4	65.1	72.9	8.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	97.7	73.2	63.8	91.2	70.9	62.0	85.5	69.2	33.7	78.4	32.0	76.0	30.5	72.8	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.0	14.9	-38.8	89.4	4.3	3.5	-41.9	91.9	-44.4	93.9	-46.8	95.8	-49.9	98.3		
(5)			-39.0	72.8	4.2	2.9	-42.0	74.9	-44.5	76.6	-46.8	78.3	-49.9	80.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.8	-5.6	-1.3	13.6	30.5	45.1	58.8	65.2	62.5	52.0	38.0	19.3	1.3
	DBStd	27.12	14.37	13.83	14.79	12.46	9.81	8.21	6.12	6.54	8.17	8.14	12.27	14.57
	HDD50	8002	1722	1438	1129	591	215	18	0	1	71	384	922	1511
	HDD65	12293	2187	1858	1594	1036	618	216	75	126	396	839	1372	1976
	CDD50	1352	0	0	0	5	63	282	470	390	132	10	0	0
	CDD65	167	0	0	0	0	1	31	79	50	6	0	0	0
	CDH74	1110	0	0	0	0	30	258	496	292	33	1	0	0
	CDH80	209	0	0	0	0	4	51	95	56	3	0	0	0
	WSAvg	7.1	6.4	6.3	6.8	6.8	7.0	6.6	6.5	7.1	7.6	8.7	8.2	6.7
	PrecAvg	22.9	0.9	0.7	1.1	1.3	1.7	2.8	3.6	3.1	2.7	2.2	1.7	1.2
(15) Precipitation	PrecMax	27.0	1.8	1.2	2.2	3.2	4.9	6.8	5.6	5.5	5.0	4.6	3.5	2.4
	PrecMin	17.2	0.4	0.3	0.2	0.3	0.4	0.8	1.2	1.6	0.9	1.0	0.8	0.5
	PrecStd	2.8	0.4	0.3	0.6	0.8	1.1	1.5	1.2	1.2	1.0	1.1	0.8	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.3	36.4	50.7	67.0	78.0	85.3	86.8	85.7	78.5	64.3	47.9	34.7
		MCWB	31.3	33.6	43.1	50.6	58.0	67.8	69.0	68.1	64.3	55.5	43.5	32.0
	2%	DB	27.3	32.4	44.0	59.7	72.2	80.1	82.5	80.7	72.7	58.0	42.2	30.0
		MCWB	26.1	30.9	37.1	46.6	55.2	63.6	66.8	66.6	61.9	50.9	38.3	28.5
	5%	DB	21.8	27.0	38.8	53.7	66.8	76.7	79.5	77.0	68.5	53.6	37.7	26.7
		MCWB	20.9	25.0	34.2	43.2	52.5	61.4	65.0	65.0	59.6	48.0	35.2	25.6
	10%	DB	16.9	20.3	34.4	48.4	61.8	73.1	76.2	73.5	64.5	49.5	34.4	22.2
		MCWB	16.1	18.7	31.4	39.6	50.0	59.7	63.5	63.3	57.5	45.5	32.9	21.3
	0.4%	WB	31.6	34.2	43.5	51.5	60.4	71.0	72.1	71.0	67.0	56.8	44.8	32.2
		MCDB	32.7	35.5	49.5	65.1	73.5	83.0	82.5	79.5	74.9	61.9	47.9	34.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.2	30.7	37.5	47.1	56.8	66.5	69.5	68.3	63.8	52.4	38.6	28.7
		MCDB	27.3	32.3	42.5	58.5	67.7	76.1	78.5	77.4	70.0	56.9	41.7	29.7
	5%	WB	21.0	25.1	34.4	43.8	54.1	64.0	67.3	66.5	60.8	48.6	35.4	25.6
		MCDB	21.9	26.9	38.3	52.8	64.1	72.4	75.6	74.5	66.8	52.8	37.7	26.7
	10%	WB	16.2	18.7	31.7	40.2	51.3	61.6	65.3	64.6	58.0	45.8	33.0	21.4
		MCDB	17.1	20.1	34.8	47.7	60.5	70.2	73.2	72.0	63.7	49.4	34.5	22.2
	5% DB	MDBR	15.9	20.3	22.2	21.3	19.6	18.0	16.4	14.9	13.3	9.5	10.4	13.7
		MCDBR	17.0	18.9	21.7	26.7	27.5	23.0	20.5	19.9	19.6	16.3	11.7	13.5
		MCWBR	16.6	17.0	16.3	16.3	14.0	10.3	8.8	9.6	10.7	10.6	9.4	12.7
		MCWBR	16.9	17.5	19.7	25.3	23.3	19.2	17.7	17.5	17.7	14.6	11.5	13.2
(35) Mean Daily Temperature Range	5% WB	MCWBR	16.6	16.0	15.0	16.1	13.1	10.0	9.1	9.5	11.3	10.2	9.6	12.5
(40) Clear-Sky Solar Irradiance	taub		0.240	0.259	0.273	0.322	0.345	0.373	0.387	0.361	0.330	0.292	0.254	0.223
		taud	2.265	2.261	2.320	2.374	2.445	2.388	2.387	2.461	2.516	2.577	2.416	2.279
	Ebn at Noon		238	270	292	289	286	278	271	273	269	259	233	222
		Edh at Noon	18	26	30	34	33	36	35	31	26	19	16	15
(44) All-Sky Solar Radiation	RadAvg		290	642	1097	1620	1710	1803	1734	1413	970	527	269	204
	RadStd		28	42	80	125	148	150	117	116	116	59	26	27

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page